

Dual-continuum models of lithium-ion batteries are fast and accurate alternatives to the Doyle-Fuller-Newman approach: I.
Derivation and validation supplementary material

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1 Introduction

In this supplementary material, we first define the remaining equations which all macroscale models (DFN, DC⁰, DC¹, and DC^y) have in common. We then give extra information on our methodology: identifying particle boundaries, calculation of C-rate, numerical choices made, convergence plots, material properties, and electrode properties. Following this, further details on the closure problem are provided. Finally, we discuss modelling options – the inclusion of macroscale diffusion, and an approximation that can be made to simplify DC¹. To maintain clear references with the equations, figures, and tables in the main text of this article, we do not reset referencing indices.

2 The Macroscale Equations in 1D for a Cathode Half Cell

2.1 External BCs for the AM Mass Balance in the DC Models

It is straightforward to model a reaction between the electrolyte within the effective separator and the AM with the DC framework; this is not the case in the DFN due to the assumption that particles are isolated. Nonetheless, modelling this reaction was found to have a negligible effect on all results reported in this work, and thus the model is defined here with “no-flux” BCs (as in the DFN) at this boundary. For the second BC, no lithium crosses the CC boundary,

$$\frac{\partial \langle c_p \rangle^p}{\partial x} = 0 \quad \text{at } x = L_s, \quad (57a)$$

$$\frac{\partial \langle c_p \rangle^p}{\partial x} = 0 \quad \text{at } x = L. \quad (57b)$$

The lengths of the separator and positive electrode are L_s and L_p respectively, and $L = L_s + L_p$.

2.2 Electrolyte Charge Balance

The PDE is,

$$\frac{\partial}{\partial x} \left(-\sigma_e \mathfrak{B}_e \frac{\partial \langle \phi_e \rangle^e}{\partial x} - \sigma_e \mathfrak{B}_e \theta \frac{\partial}{\partial x} \ln \langle c_e \rangle^e \right) = a_k j_k^m \quad \text{for } 0 < x < L. \quad (58)$$

where a_k is the specific surface area ($\frac{A}{V}$), j_k^m depends on the model being solved, with $m \in \{0, 1, y, d\}$ and,

$$\mathfrak{B}_e = \begin{cases} \mathfrak{B}_{e,s} & \text{for } 0 \leq x \leq L_s, \\ \mathfrak{B}_{e,p} & \text{for } L_s < x \leq L, \end{cases} \quad a_k = \begin{cases} 0 & \text{for } 0 \leq x \leq L_s, \\ a_p & \text{for } L_s < x \leq L, \end{cases} \quad j_k^m = \begin{cases} 0 & \text{for } 0 \leq x \leq L_s, \\ j_p^m & \text{for } L_s < x \leq L, \end{cases} \quad (59)$$

where $\mathfrak{B}_e$ is the transport efficiency of the electrolyte phase, which is related to the porosity (ε_e) and tortuosity (τ_e) by,

$$\mathfrak{B}_e = \frac{\varepsilon_e}{\tau_e}. \quad (60)$$

The BV expression for reaction rate at a lithium foil is,

$$j_{li} = 2j_{li,0} \sinh \left(-\frac{F}{2RT} \langle \phi_e \rangle^e \right). \quad (61)$$

For a constant-current discharge, the current density provided by the cell (i_{app}) will determine the current density at the lithium foil due to charge conservation. This expression can then be rearranged

to find a Dirichlet BC for $\phi_{e,s}$. However, note that the surface porosity of the effective separator, $\varepsilon_{e,s}^{\text{surf}}$ must be accounted for in this,

$$A_{cc}i_{\text{app}} = - \int_{\mathcal{A}_s} \varepsilon_{e,s}^{\text{surf}} j_{li} dA, \quad (62)$$

where $\mathcal{A}_s$ is the set including all points on the lithium foil – effective separator interface, and A_{cc} is the area of the current collector. Since j_{li} is constant across $\mathcal{A}_s$ and $\int_{\mathcal{A}_s} dA = A_{cc}$,

$$j_{li} = -\frac{i_{\text{app}}}{\varepsilon_{e,s}^{\text{surf}}}, \quad (63)$$

and thus,

$$\langle \phi_e \rangle^e = -\frac{2RT}{F} \text{arcsinh} \left(-\frac{i_{\text{app}}}{2\varepsilon_{e,s}^{\text{surf}} j_{li,0}} \right) \quad \text{at } x = 0. \quad (64)$$

For the BC at $x = L$, the CC is assumed to be impermeable to lithium ions. Using the expression for $\langle i_e \rangle^e$ with this statement we obtain,

$$\frac{\partial \langle \phi_e \rangle^e}{\partial x} = 0 \quad \text{at } x = L. \quad (65)$$

2.3 Electrolyte mass balance

The PDE is,

$$\varepsilon_e \partial_t \langle c_e \rangle^e = \frac{\partial}{\partial x} \left(D_e \mathfrak{B}_e \frac{\partial \langle c_e \rangle^e}{\partial x} \right) + F^{-1} (1 - t_+^0) a_k j_k^m \quad \text{for } 0 < x < L, \quad (66)$$

where the electrolyte volume fraction (ε_e) is

$$\varepsilon_e = \begin{cases} \varepsilon_{e,s} & \text{for } 0 \leq x \leq L_s, \\ \varepsilon_{e,p} & \text{for } L_s < x \leq L. \end{cases} \quad (67)$$

At the lithium-foil boundary with the effective separator, the macroscale electrolyte current ($\bar{i}_{e,s}$) and flux ($\bar{N}_{e,s}$) must balance the charge entering/leaving the cell at the CC,

$$i_{\text{app}} = \bar{i}_{e,s} \quad \text{at } x = 0, \quad (68a)$$

$$\frac{i_{\text{app}}}{F} = \bar{N}_{e,s} \quad \text{at } x = 0. \quad (68b)$$

The expression for $\bar{N}_e$,

$$\bar{N}_{e,s} = -\mathfrak{B}_{e,s} D_e \frac{\partial \langle c_e \rangle^e}{\partial x} + \frac{t_+^0}{F} \bar{i}_e, \quad (69)$$

can be used in Equation 68 to give

$$\frac{\partial \langle c_e \rangle^e}{\partial x} = -\frac{1}{F \mathfrak{B}_{e,s} D_e} (1 - t_+^0) i_{\text{app}} \quad \text{at } x = 0. \quad (70)$$

For the BC at $x = L$, the CC is assumed to be impermeable to lithium ions,

$$\frac{\partial \langle c_e \rangle^e}{\partial x} = 0 \quad \text{at } x = L. \quad (71)$$

Finally, we require in initial condition,

$$\langle c_e \rangle^e = c_{e,0} \quad \text{at } t = 0. \quad (72)$$

2.4 AM Charge Balance

The PDE is,

$$\frac{\partial}{\partial x} \left(\sigma_k \mathfrak{B}_k \frac{\partial \langle \phi_k \rangle^k}{\partial x} \right) = a_k j_k^m \quad \text{for } L_s < x < L. \quad (73)$$

where $\mathfrak{B}_k$ is the transport efficiency of the AM phase, which is related to AM volume fraction and tortuosity by,

$$\mathfrak{B}_k = \frac{\varepsilon_k}{\tau_k}. \quad (74)$$

An effective conductivity can also be defined as,

$$\sigma_k^{\text{eff}} = \sigma_k \mathfrak{B}_k. \quad (75)$$

For a galvanostatic (dis)charge, the current within the AM is fixed by the applied current density,

$$\frac{\partial \langle \phi_p \rangle^p}{\partial x} = -\frac{i_{\text{app}}}{\mathfrak{B}_k \sigma_p} \quad \text{at } x = L. \quad (76)$$

We assume that no reaction takes place at the interface between the electrode and the effective separator, giving

$$\frac{\partial \langle \phi_p \rangle^p}{\partial x} = 0 \quad \text{at } x = L_s. \quad (77)$$

3 Identifying Particle Boundaries and Defining Length Scales

For every macroscale model, particle boundaries must be defined for the electrode image to complete parameterisation. For DC¹, the particle label map is used to solve the closure problem, whilst for DC^y and DFN, it is used to define a length scale. A watershed algorithm was used to identify particle boundaries [38, 63] and define a particle label map. The process can be summarised as follows:

1. Scipy's Euclidean distance transform was used on the binary electrode image to calculate the distance of AM cells from the electrolyte interface, $\partial\Omega_{k,e}$.
2. Local maxima of the distance map were computed to serve as seed points.
3. These seed points were then used to guide the watershed segmentation, which simulates the flooding of basins from the seed points. Boundaries are formed where growing regions meet, partitioning the continuous electrode into particles.

The length scale of each particle can be calculated by taking the maximum value of the distance map from Step 1 for each particle region.

4 Calculation of C-Rate

In this work, the C-rate is defined based on the theoretical half-cell capacity, Q_T ,

$$Q_{T,k} = FV_k c_{k,\text{max}}. \quad (78)$$

Therefore, the applied current density resulting from a certain C-rate (C) is

$$i_{\text{app}} = \frac{Q_{T,k} C}{3600 A_{cc}}. \quad (79)$$

Table 5: The computation time for each model

Electrode	DFN (s)	DC ⁰ (s)	DC ¹ (s)	DC ^y (s)
A	0.278	0.0601	0.132	0.124
B	0.285	0.0607	0.0733	0.0701
C	0.284	0.0592	0.0744	0.0727
D	0.341	0.0639	0.0714	0.0730
3D	0.288	0.0645	0.0805	0.0739

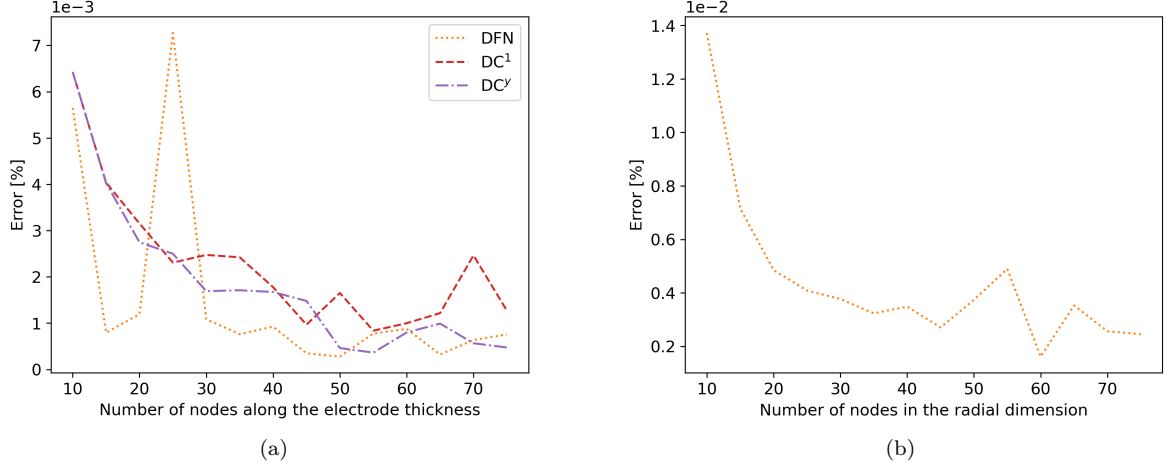


Figure 9: The error against number of nodes along the electrode thickness (a), with 60 nodes in the radial dimension for the DFN. The error against the number of nodes in the radial dimension of the DFN model (b), with 40 nodes along the electrode thickness.

5 Solver Parameters

For the macroscale models, a discretisation of 20 nodes in the separator, 40 nodes in the positive electrode, and 60 nodes in the radial direction (exclusively for the DFN) was used. PyBaMM's IDAKLU solver was used with default parameters to solve all models. Computations were performed using a single core on a Windows Subsystem for Linux system with a 12th Gen Intel(R) Core(TM) i7-12700H 2.30 GHz processor and 32 GB of RAM available. The computation times for each model and each case are shown in Table 5. The solution convergence against the number of nodes used is shown in Figure 9. To evaluate this, the simulation on Electrode 3D was used, and the integral of the cell voltage over the discharge time is used as a metric to calculate a percentage error,

$$\text{Error} = \frac{|\int_0^{t_d} V_i dt - \int_0^{t_d} V_{\text{ref}} dt|}{\int_0^{t_d} V_{\text{ref}} dt} \times 100, \quad (80)$$

where V_i is the cell-voltage solution with i nodes, and V_{ref} is the reference cell-voltage solution using 500 nodes. The times at the end of the discharge, $t_{d,i}$ and $t_{d,\text{ref}}$, are named analogously. This is more appropriate than some form of root-mean squared error, because $t_{d,i}$ and $t_{d,\text{ref}}$ may not be equal, thus there is a region where $V_i - V_{\text{ref}}$ is not defined.

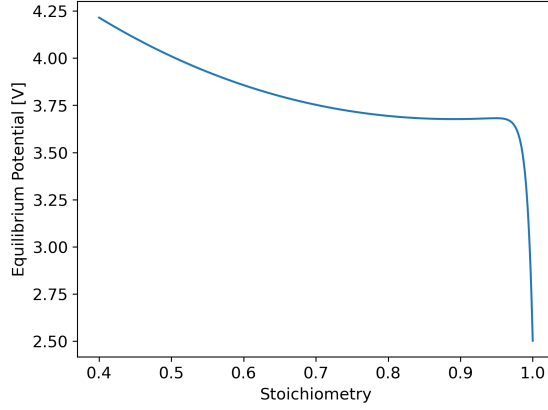


Figure 10: The equilibrium potential as a function of stoichiometry [34]

Table 6: Mean stoichiometry at the end of the discharge for the simulations shown in Figures 3 and 5a

Model	Fig. 3a	Fig. 3b	Fig. 3c	Fig. 3d	Fig. 5a
DC ⁰	0.9932	0.9938	0.9936	0.9937	0.9935
DC ^y	0.9628	0.9892	0.9781	0.9923	0.9837
DC ¹	0.9582	0.9806	0.9750	0.9904	0.9796
DFN	0.9822	0.9926	0.9874	0.9935	0.9881
DNS	0.9607	0.9798	0.9645	0.9857	0.9799

6 The OCP Function

The OCP function used for NMC, $U_k(x)$, is shown in Figure 10. The equation was taken from Hein's work [34],

$$U_k = 13.49050 - 10.96038x + 8.203617x^{1.358699} - (3.10758 \times 10^{-6})e^{127.1216x - 114.2593} - 7.033556x^{-3.362749 \times 10^{-2}}, \quad (81)$$

where x is the stoichiometry, defined as,

$$x = \frac{c_k}{c_{k,\max}}. \quad (82)$$

7 Stoichiometry at the end of Discharge

To aid analysis of the results in Figures 3, 5a, and 6, the mean stoichiometries at the end of the discharge (when the voltage constraint is met) are shown for each model in Tables 6 and 7. The mean stoichiometry, $\bar{x}$, is defined as,

$$\bar{x} = \frac{1}{V_k c_{k,\max}} \int_{\Omega_k} c_k dV. \quad (83)$$

Table 7: Mean stoichiometry at the end of the discharge for the simulations shown in Figure 6

Model	Fig. 6a	Fig. 6b
DC ^y	0.9716	0.9533
DC ¹	0.9631	0.9117
DFN	0.9764	0.9702
DNS	0.9564	0.8953

8 Additional Simulations with Non-Constant Material Properties

8.1 Introduction

As mentioned in the main text, one potential criticism of our results is the fact that constant values were used for electrolyte and AM material properties, whilst these can depend strongly on c_e and c_k respectively. We consider constant values a good starting point, as numerous articles in the literature employ this approach for the electrolyte and/or the AM [28, 34–36, 60, 61], and they demonstrate the performance of the models whilst minimising complexity. Nonetheless, to confirm that the use of non-constant material properties does not alter our conclusions, an extra case is included here, in which D_e , σ_e , and $(1 + \delta_e)$ are functions of c_e , and D_k is a function of c_k .

8.2 Methodology

The electrolyte property functions have been taken from Valoen’s work [58], whilst the expression for $D_k(c_k)$ was published by Schmalstieg et al. [64]; these are shown in Figure 11, and all other material properties are unchanged. Since the transference number was found to not vary significantly with c_e in Valoen’s work, we have kept it as a constant value. Similarly to the results in Figure 6, Electrode B was used, and a 1C discharge to 3.5 V followed by a 15 minute relaxation was simulated.

Using $D_k(c_k)$ has some impact on our upscaling approach. During the volume-averaging process, we assumed that D_k does not vary significantly over the scale of an averaging volume (see Equation 17). Of course, using $D_k(c_k)$ will degrade the validity of this assumption, but we maintain this approach nonetheless – one could develop an extension of the model to include the effects of D_k variations across the averaging volume. When parameterising DC¹ with $D_k(c_k)$, the dimensionless form of the closure problem should be solved, to avoid its solution depending on c_k .

8.3 Results

The cell-voltage results are shown in Figure 12a. Here, we again see that DC¹ is able to capture the final discharge time more precisely than DC^y and DFN. Likewise, throughout the discharge, DC¹ provides the most accurate cell-voltage predictions.

Due to the relatively low values of D_k in this case (see Figure 11d), AM diffusion is more limiting than it was for the simulations using $D_k = 4 \times 10^{-14} \text{ m}^2\text{s}^{-1}$ presented in the main article. Therefore, as observed earlier in Figure 6b, the differences between the different models – most notably the final discharge times – are accentuated, since the treatment of AM mass transport becomes more influential. Likewise, this larger diffusion limitation causes a more significant error in DC¹’s prediction at the start of the discharge: as discussed in the main article, an extension presented in Part II is able to resolve this problem.

Initially it appears that DC¹ does not capture the voltage plateau during the relaxation period as well as DC^y and DFN. However, upon closer inspection, we see that the very low value of D_k at

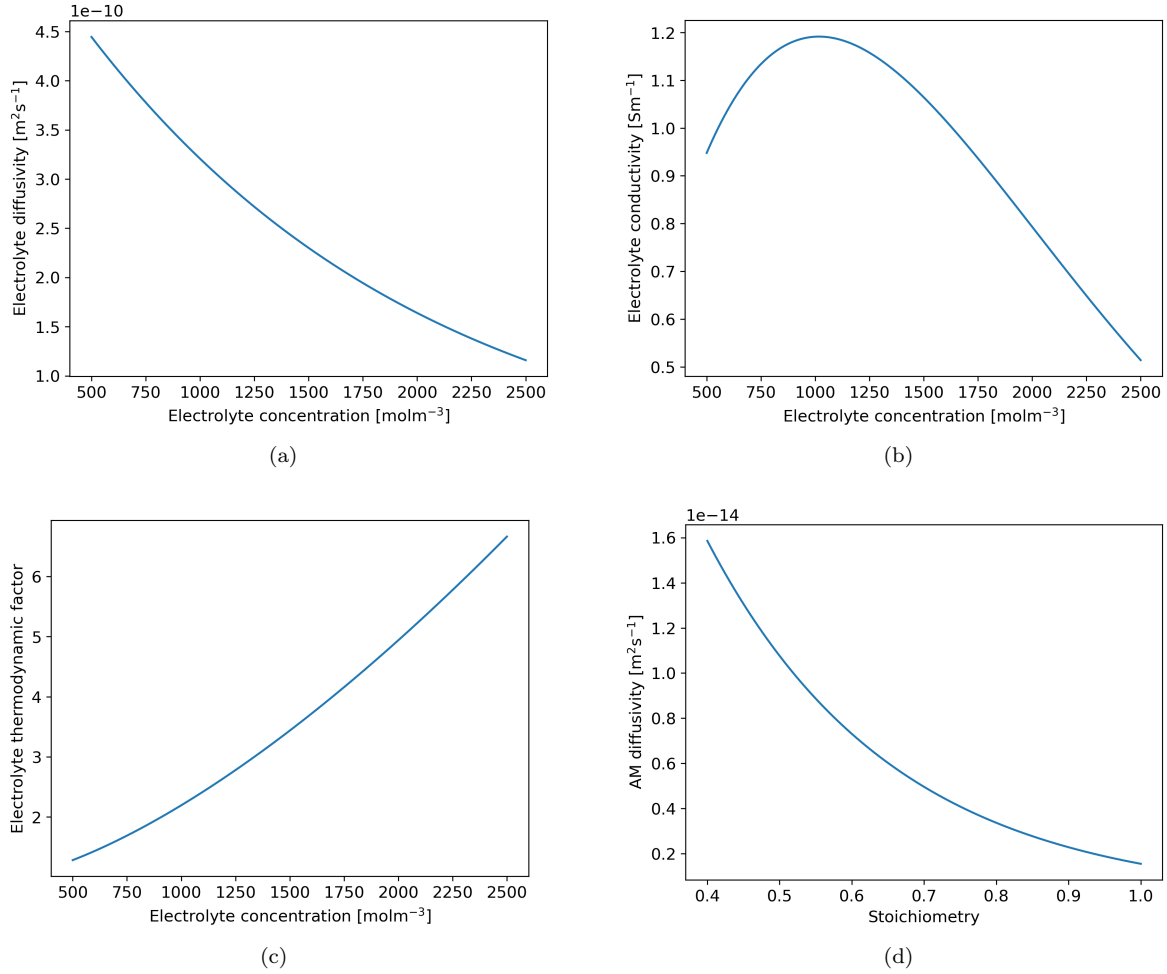


Figure 11: D_e (a), σ_e (b), and $(1 + \delta_e)$ (c) as functions of c_e [58], and D_k as a function of stoichiometry (d) [64]

high stoichiometry causes an extremely slow relaxation. If the length of the relaxation is extended, as shown in Figure 12b, we see that DC¹ does indeed predict the final voltage plateau rather well.

To summarise, we argue that these results support the claim that our conclusions (most importantly regarding the performance of DC¹) will still be valid when material properties are non-linear functions of state variables.

9 Additional Details on the Closure Problems

Two options were presented for solving the closure problem, with (Option 1) and without (Option 2) interparticle flux. A comparison of $\langle s \rangle_A$ for the two methods is shown in Table 8. For the method used in the main article (Option 1), the computation time of the closure problem is shown in Table 9. Note that this is the total time taken to perform all operations necessary in OpenFOAM, rather than just the time for the solver to complete – mesh building and result field reconstruction can constitute a substantial portion of the time. We also highlight that this computation time could likely be reduced significantly by optimising the OpenFOAM parallelisation, workflow and solver choices – this has not been pursued yet. Finally, the closure problem can be non-dimensionalised using the following

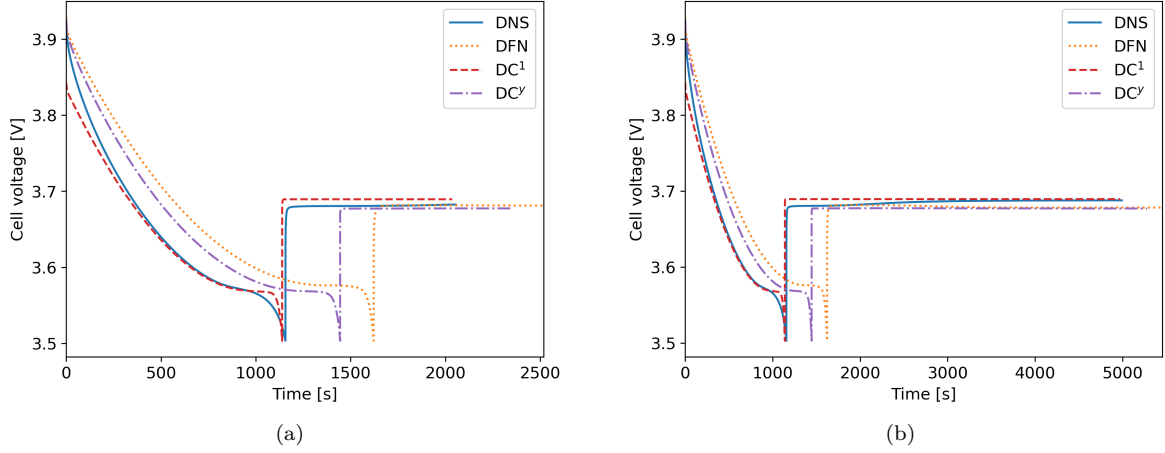


Figure 12: Cell-voltage profiles for Electrode B when using non-constant material properties (a), with an extended relaxation period to show the slow relaxation of the DNS (b)

Table 8: $\langle s \rangle_A$ for Options 1 and 2

Parameter	Option 1 $\langle s \rangle_A$ with interparticle flux	Option 2 $\langle s \rangle_A$ without interparticle flux
A	-492.300	-491.996
B	-451.363	-453.900
C	-387.966	-387.966
D	-190.679	-162.239
3D	-416.708	-532.659

dimensionless groups,

$$s_i = \frac{L}{D_k F} \hat{s}_i, \quad (84a)$$

$$x = \hat{x} L. \quad (84b)$$

For Option 1, this gives

$$\hat{\Delta} \hat{s}_i = -\frac{L A_i}{V_{k_i}} \quad \text{in } \Omega_{k_i}, \quad (85a)$$

$$\mathbf{n} \cdot \hat{\nabla} \hat{s}_i = -1 \quad \text{at } \partial \Omega_{k_i, e}, \quad (85b)$$

$$\mathbf{n} \cdot \hat{\nabla} \hat{s}_i = \mathbf{n} \cdot \hat{\nabla} \hat{s}_j \quad \text{at } \partial \Omega_{k_i, k_j}, \quad (85c)$$

$$\hat{s}_i = \hat{s}_j \quad \text{at } \partial \Omega_{k_i, k_j}. \quad (85d)$$

10 Developing DC¹ Without Neglecting Macroscale Diffusion

Here we demonstrate that DC¹ can easily be developed without the assumption in Equation 21. Nonetheless, we found that the more complicated model obtained offered a negligible improvement

Table 9: The computation time for the closure problem of each electrode

Parameter	Number of processors	Computation time
A	1	22.0 s
B	1	30.8 s
C	8	80.1 s
D	8	26.3 mins
3D	64	11.7 hrs

compared to the original model. Returning to the PDE and BCs on $\tilde{c}_{k_i}$ (Equations 24, 25 and 32) but avoiding this assumption, we instead have

$$D_k \Delta \tilde{c}_{k_i} = -\frac{A_i}{FV_{k_i}} \langle j_{k_i} \rangle_A \quad \text{in } \Omega_{k_i}, \quad (86a)$$

$$-\mathbf{n} \cdot D_k \nabla (\tilde{c}_{k_i} + \langle c_{k_i} \rangle^k) = \frac{\langle j_{k_i} \rangle_A}{F} \quad \text{at } \partial\Omega_{k_i,e}, \quad (86b)$$

$$\mathbf{n} \cdot D_k \nabla (\tilde{c}_{k_i} + \langle c_{k_i} \rangle^k) = 0 \quad \text{at } \partial\Omega_{k_i,k_j}. \quad (86c)$$

We show Option 2 for the treatment of $\partial\Omega_{k_i,k_j}$, as it simplifies the analysis. An extra macroscale source term appears, $\nabla \langle c_{k_i} \rangle^k$, leading us to propose the following closure,

$$\tilde{c}_{k_i} = \mathbf{b}_i \cdot \nabla \langle c_{k_i} \rangle^k + s_i \langle j_{k_i} \rangle_A. \quad (87)$$

By inserting this into Equation 86 and exploiting linearity, we arrive at separate problems for $\mathbf{b}_i$ and s_i . The problem on s_i is identical to that shown above (Equation 34), whilst the problem for $\mathbf{b}_i$ is

$$\Delta \mathbf{b}_i = 0 \quad \text{in } \Omega_{k_i}, \quad (88a)$$

$$-\mathbf{n} \cdot \nabla \mathbf{b}_i = \mathbf{n} \quad \text{at } \partial\Omega_{k_i,e}, \quad (88b)$$

$$-\mathbf{n} \cdot \nabla \mathbf{b}_i = \mathbf{n} \quad \text{at } \partial\Omega_{k_i,k_j}. \quad (88c)$$

This closure problem is classical for diffusion in porous media [7]. Returning to the PDE on $\langle c_{k_i} \rangle^k$ before macroscale diffusion was neglected (Equation 20), inserting the expression for $\tilde{c}_{k_i}$, and neglecting macroscale gradients in $\langle j_{k_i} \rangle_A$ gives

$$\partial_t \langle c_{k_i} \rangle^k = \nabla \cdot D_k \left(\mathbf{I} + \frac{1}{V_{k_i}} \int_{\mathcal{A}_i} \mathbf{n} \mathbf{b}_i dA \right) \cdot \nabla \langle c_{k_i} \rangle^k - \frac{A_i}{FV_{k_i}} \langle j_{k_i} \rangle_A. \quad (89)$$

The term $\left(\mathbf{I} + \frac{1}{V_{k_i}} \int_{\mathcal{A}_i} \mathbf{n} \mathbf{b}_i dA \right)$ is effectively a tortuosity and corrects the diffusion coefficient.

11 Approximating the Reaction Rate in DC¹

In the development of DC¹, we used the approximation that $\langle j_k \rangle_A \approx j_k^1$ in Equation 44, since we do not know $\langle j_k \rangle_A$ *a priori*. However, a good estimate of $\langle j_k \rangle_A$ can generally be obtained by assuming that the reaction rate is evenly distributed across the electrode surface,

$$\langle j_k \rangle_A^c = \lambda_k \frac{i_{\text{app}}}{L_k a_k}. \quad (90)$$

where i_{app} is the current density applied to the current collector, $\lambda_n = -1$ and $\lambda_p = 1$. We can therefore use the following expression for $\langle c_k \rangle_A^1$ when calculating j_k^1 ,

$$\langle c_k \rangle_A^1 = \langle c_k \rangle^k + \langle j_k \rangle_A^c \langle s \rangle_A. \quad (91)$$

Using such an approach, the correction term $\langle j_k \rangle_A^c \langle s \rangle_A$ can be calculated *a priori*. In all simulations shown in this work, using this assumption introduced a negligible error, and slightly reduced the computation time for the model.